

# Technologies + Concepts

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## Data Processing

- Pyresparser + spaCy → Resume se data nikalne ke liye
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### Pyresparser

- Pre-built resume parser library
- Already trained for resume fields

#### Kya nikalta hai?

- Name
- Email
- Skills
- Experience
- Education

#### Pros:

- Ready-made solution
- Fast
- Less coding
- Resume-specific

#### Cons:

- Limited customization

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### spaCy

- Powerful NLP library
- Used for:
  - o Named Entity Recognition (NER)
  - o POS tagging

#### Pros:

- High accuracy
- Custom models bana sakte ho
- Industry-level

#### Cons:

- Thoda complex
- Setup + training required

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# Keyword Extraction

**NLP / RNN / LLaMA → Important skills identify karne ke liye**

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## NLP

Basic keyword extraction  
Text ko tokenize karke important words nikalta hai

**Pros:**

Easy  
Fast

**Cons:**

Context samajhne me weak

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## RNN

- Sequence-based deep learning model
- Words ke relation ko samajhta hai

**Pros:**

Better accuracy  
Context samajhta hai

**Cons:**

Training required  
Time-consuming

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## LLaMA (LLM)

- Advanced language model
- Prompt-based intelligent extraction

**Pros:**

Best accuracy  
Context + meaning samajhta hai

**Cons:**

Heavy model  
Resource cost

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## Matching

- **Embeddings + Cosine Similarity → Resume vs Job match**

- Text ko numerical vectors me convert karta hai
- Resume aur job description ka similarity score nikalta hai

**Output:**

- Match % (e.g., 85%)
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## Intelligence / Suggestions

- **LLM (GPT) → Suggestions + resume improvement**

- Resume analyze karta hai
  - Improvement suggestions deta hai
  - Better sentences generate karta hai
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## Recommendation

- **ML + Rule-based system → Missing skills & courses**

- User skills vs required skills compare karta hai
  - Missing skills detect karta hai
  - Courses suggest karta hai
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## Storage

- **Database / Vector DB → Resume & embeddings store karne ke liye**

- Resume data store hota hai
  - Embeddings store hote hain
  - Fast retrieval ke liye use hota hai
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## Complete End-to-End Flow

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## STEP 1: Resume Upload

### User:

- PDF / DOC file upload karta hai

### System:

- File validate karta hai (format, size)
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## STEP 2: Resume Parsing (NLP)

- Resume ko structured data me convert karte hain

### Extract hota hai:

- Name
- Skills
- Education
- Experience

### Tools:

- spaCy / Pyresparser
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## STEP 3: Text Cleaning & Preprocessing

- Unnecessary words remove
- Stopwords remove
- Text normalize

### Result:

- Clean resume text
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## STEP 4: Keyword Extraction

### Goal:

- Important skills detect karna

### Methods:

- NLP (basic)

- RNN (deep learning)
- LLaMA (LLM-based)

**Output:**

- Skills list (Python, ML, SQL)
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## STEP 5: Resume vs Job Matching

- Resume aur job description compare

**Process:**

- Text → embeddings
- Similarity calculate

**Output:**

- Match score (e.g., 85%)
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## STEP 6: Skill Gap Analysis

- Required vs user skills compare

**Output:**

- Missing skills list

**Example:**

- User → Python
  - Required → Python, ML, DL
  - Missing → ML, DL
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## STEP 7: Job Role Prediction

- Best job role suggest

**Example:**

- Skills → Python + ML
- Job → Data Scientist

**Use:**

- ML model / LLM

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## STEP 8: Resume Scoring

- Skills
- Projects
- Experience
- Certifications
- Format

### Output:

- Score (0–100)
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## STEP 9: Recommendation System

System suggest karta hai:

### Skills:

- Missing skills

### Courses:

- Udemy / Coursera

### Resume Tips:

- Add projects
- Improve summary
- Add achievements

### Use:

- ML + LLM
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## STEP 10: Resume Improvement (LLM)

### GPT:

- Resume analyze karta hai
- Better version suggest karta hai

### Example:

- Weak line → strong professional line
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## STEP 11: Output Dashboard

**User ko milta hai:**

- Resume score
- Job match %
- Missing skills
- Recommendations
- Improved suggestions

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# FINAL FLOW

